Sets & Relation

Mind map

Sets
- Roster / set-builder
- Finite, infinite, singleton, ϕ
- Equal vs equivalent

Subsets
- 2^n subsets, $2^n - 1$ proper
- Superset $B \supseteq A$
- $n(P(A)) = 2^n$

Operations
- $\cup,\ \cap,\ A - B = A \cap B'$
- $A \,\Delta\, B$, complement
- Disjoint: $A \cap B = \phi$

Laws
- Distributive laws
- $(A \cup B)' = A' \cap B'$
- Venn: 8 regions for 3 sets

Counting elements
- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- Three-set formula, exactly one / two
- Bounds on $n(A \cap B)$
- Multiples: $[N/k]$, lcm

Cartesian product
- $n(A \times B) = mn$
- Distributes over $\cup, \cap$
- Common part $n(A \cap B)^2$

Relation
- $R \subseteq A \times B$
- Domain, co-domain, range
- 2^{mn} relations

Properties
- Reflexive: $y = x$ test
- Symmetric: swap test
- Transitive: chain / counter-example
- Anti-symmetric

Equivalence
- R + S + T
- Classes form a partition
- Bell numbers $B_3 = 5,\ B_4 = 15$

Counting relations
- Ref $2^{n(n-1)}$, Sym $2^{n(n+1)/2}$
- Minimum pairs to add
- Count by fixing first entry

Inverse relation
- $R^{-1} = \{(b, a)\}$
- Dom and range swap
- Symmetric $\iff R = R^{-1}$